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AI FOR LOST CHILD IDENTIFICATION

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ABSTRACT

Children's safety has been a very important issue of concern for citizens, particularly at crowded public places such as malls and parks where children often get lost. Traditional monitoring systems where multiple CCTVs watch every moving person have problems like human eye fatigue, delay in giving responses, and even mistakes made by the humans. That is why we have developed a computer-based system to identify lost children automatically using a set of smart algorithms, known as Lost Children Identification System (LCIS). LCIS system is capable of detecting lost children with a high degree of accuracy by making use of computer vision technology. The system utilizes MTCNN for accurate face detection and alignment. FaceNet is employed to generate 512-dimensional facial embeddings, which are compared with a prestored database using cosine similarity. A dual threshold decision mechanism is applied to improve accuracy and reduce false positives. Upon detecting a high-confidence match, the system identifies the child and provides the corresponding similarity score. The proposed solution reduces dependence on manual monitoring and offers a scalable and efficient approach for real-time child identification in surveillance environments.

Keywords: Artificial Intelligence, Machine Learning, Deep Learning, Face Detection, Face Recognition, Image Processing.

I. INTRODUCTION

Face recognition has become one of the most important applications of Artificial Intelligence in recent years. It is widely used in security systems, surveillance, mobile authentication, and identity verification. A face recognition system generally consists of two main stages: face detection and face recognition. Face detection identifies the presence and location of a face in an image, while face recognition verifies or identifies the person. Several researchers have proposed different techniques for improving face detection performance. The Multi-Task Cascaded Convolutional Neural Network (MTCNN) has shown strong performance in detecting faces under varying lighting conditions, poses, and facial expressions. MTCNN is a facial detection system made up of three components: P-Net, R-Net, and O-Net. The three components collaboratively work in identifying a face and refining the facial points. This, in turn, helps increase the accuracy of detection and reduces false positives. Besides, MTCNN can handle cases of data scarcity or data imbalance. Researchers who are passionate about MTCNN have discovered how to enhance the performance of the MTCNN method. For instance, in 2025, Wang and Liu showed a new MTCNN variant featuring the so-called Ghost module. Without compromise of the MTCNN's accuracy, this upgraded version of MTCNN is less resource-intensive to operate. That said, the original MTCNN implementation is widely used since it is straightforward, dependable, and augments well with MTCNN. Other face retrieval systems such as FaceNet have contributed to raising the bar. FaceNet basically transforms image files into 512-dimensioned vector embeddings. Such numeric representation allows FaceNet to gather the images of one individual into one cluster. Several research works highlight the superiority of FaceNet over baseline face recognition approaches. The situation is especially dramatic with different illumination conditions, facial expressions or side views when using FaceNet. This investigation is about developing an intelligent identification system for locating missing children. It is based on MTCNN for face detection and FaceNet for identity confirmation. The face pair is compared based on the distance and the decision is made if the faces belong to the same person based on the threshold present in MTCNN and FaceNet. Combining MTCNN and FaceNet allows the identification of children. As an approach, it is very simple, at the same time accurate and practical for implementation in real-world scenarios where people surveillance is needed and MTCNN and FaceNet are the tools of choice.

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II. METHODOLOGY

The proposed system automatically identifies missing or lost children by using deep learning-based face recognition. The main objective is to lessen the manual task in surveillance systems, enhance the pace and accuracy of recognizing children in crowded public spaces like malls, railway stations and events.

The working process of the system as a whole is classified into different stages which are further mentioned in a simple manner:

Data Collection & Dataset Preparation

The first part of the system is that the dataset is to be prepared first. Images of children will be taken and then arranged systematically so that each child can be represented by a folder containing multiple images.

The images collected will be under various conditions:

Different light sources

Different facial expressions

Slight pose variations (right, left, front etc.)

The dataset will be then cleaned before it can be trained or used:

Corrupt and blurry images removed

All images will only have one face in each image

All images converted to same format (JPG/PNG)

Now the dataset is ready for feature extraction.

Face Detection with MTCNN

Once an image or a video frame is given to the system it has to detect the face first.

This will be done using MTCNN (Multi-task Cascaded Convolutional Neural Network).

This technique works in 3 stages:

a) P-Net (Proposal Network):

Scans the image rapidly and detects possible face regions.

B) R-Net (Refinement Network):

Filters false positives and detects facial boundaries.

C) O-Net (Output Network):

Identifies the final location and extracts landmarks of the faces (eyes, nose and mouth).

Thus, even in crowd and poor quality image, the correct face is processed.

Feature Extraction using FaceNet

Once the face has been detected the next process is to extract the essential information from the face. This will be done using the FaceNet deep learning model (Inception-ResNet V1).

FaceNet maps the detected face to a 512-dimensional vector or embedding.

This embedding would encode information such as:

Unique structure of the face, the distance between different facial features, data about the identity in numerical form.

In order to get accurate results from comparing these embeddings L2 normalization would be done on each vector

Database Creation using Centroid Embeddings

When multiple images are present for a particular child a centroid or average embedding for that child will be calculated and the same will be stored rather than all the embeddings separately. This is calculated by:

Summing all the embeddings for one child and finding the mean of them.

This single centroid acts as the identity point for that child.

This reduces space as compared to all the embeddings separately and hence making matching quicker and the identity more stable.

Face Matching using Cosine Similarity

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Once a new face is presented to the system its embedding will be compared with all the child centroid embeddings in the database using cosine similarity.

The cosine similarity calculates how close the two face vectors are to each other, if it's very close both are the same faces and vice versa. Since all embeddings are normalized, cosine similarity is the same as performing a dot product of the vectors.

Dual Threshold Decision Mechanism

A dual-threshold mechanism is proposed to improve the recognition rate and reduce incorrect classifications. A face will be marked as match if:

Score is greater than T1 (minimum threshold)

The difference between top 2 matches is greater than T2 (margin threshold)

If both conditions are met the face will be marked as a match or if not "Unknown" label will be displayed.

This will make the system efficient in practical conditions such as:

Identical-looking children

Bad lighting conditions

Obscured face.

Real-Time Video Processing

For live surveillance each video frame will be processed independently through the MTCNN and FaceNet models and matched against the database of children.

Each frame will go through:

Face detection (MTCNN)

Feature extraction (FaceNet)

Matching with database

Showing the matched identity and image in the same frame.

The output video would thus display the bounding boxes around the faces with a label with the children's ID/name and their match score.

System Output and Alert Generation

Upon matching faces a successful ID for a missing child will be shown. An alert can also be generated when a match is found, as per the system requirements during deployment.

III. MODELING

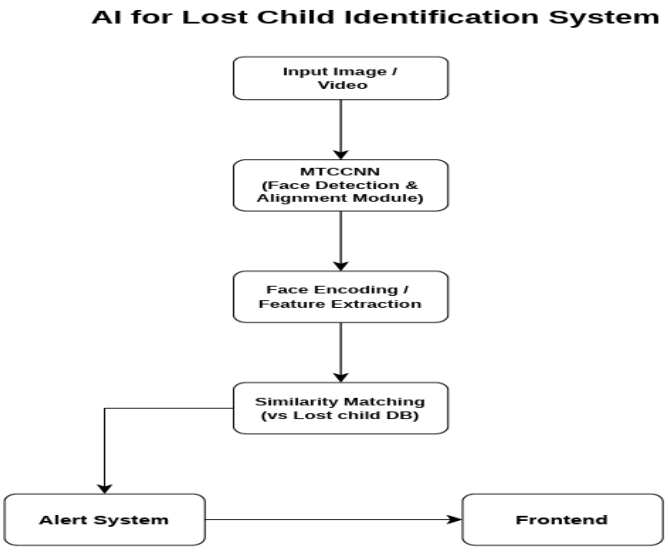


Figure 1: Proposed System Architecture

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IV. RESULTS

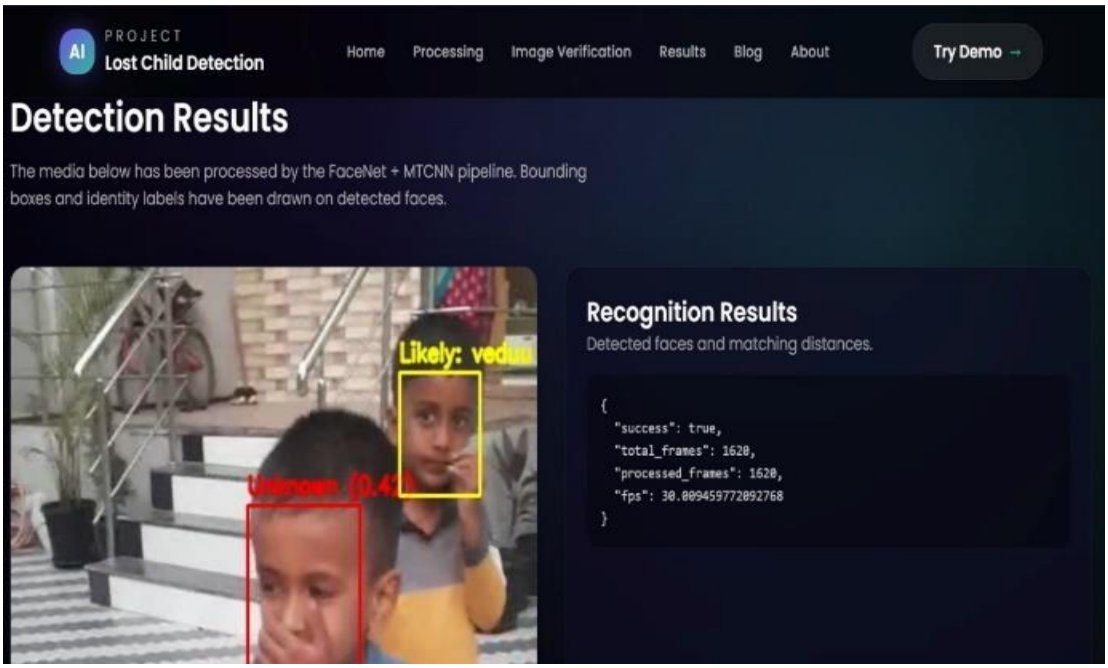


Figure 2: Detection results of AI Lost Child System

V. CONCLUSION

The Lost Child Identification System is a system that uses computer methods to find and recognize faces. It uses something called MTCNN to find faces and FaceNet to create a kind of map of each face. This map is like a fingerprint for each persons face. The system then uses these maps to find matching faces by measuring how similar they are. It does this by using something called cosine similarity and a special way of deciding if two faces match. The Lost Child Identification System also uses something called YOLO to find objects in pictures and videos. This helps the system work better when it is used to watch people in places. The Lost Child Identification System can. Recognize children quickly and accurately from pictures and videos. When we tested the Lost Child Identification System it worked well and did not make many mistakes. The Lost Child Identification System uses a way of classifying faces that makes it more consistent and reliable. When we looked at the results, we saw that the Lost Child Identification System was very good at recognizing faces. The Lost Child Identification System is a way to identify lost children and it could also be used to help keep people safe in public places. The Lost Child Identification System is a solution for finding lost children. The Lost Child Identification System works well. Is reliable. The Lost Child Identification System could also be used for things like watching people, in public places and keeping them safe.

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